

The schematic diagram illustrates the internal structure and pin connections of the Raspberry Pi RP2350A QFN-60 package. The chip is shown in a yellow box with its internal components and connections. The power supply section includes a +3V3 rail with a 33R resistor (R5) and a 4u7 capacitor (C9), and a +1V1 rail with a 100nF capacitor (C11) and a 4u7 capacitor (C7). The ground connections are shown as green lines. The signal and control pins are connected to various components, including a 12MHz crystal oscillator (Y1) and a 1k resistor (R2). The peripheral connections include USB_DP, USB_DM, and various GPIO pins. The diagram is color-coded: red for power supply and ground connections, green for signal and control connections, and blue for peripheral connections. The pins are numbered 1 to 60, with some pins having specific functions like GPIO, I2C, SPI, and USB.

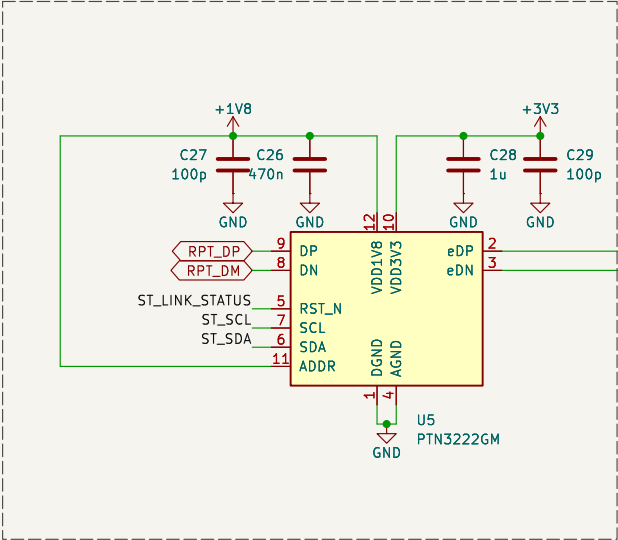
The diagram shows two switches, SW1 and SW2, connected to the BOOTSEL and RUN pins of a microcontroller. SW1 is a momentary switch connected to GND and BOOTSEL. SW2 is a momentary switch connected to a 1k resistor (R4) and RUN, with the other end of the resistor connected to GND.

The diagram shows two pin headers, J2 (LEFT) and J3 (RIGHT), with their respective pin connections. J2 has 16 pins, and J3 has 16 pins. The connections are as follows:

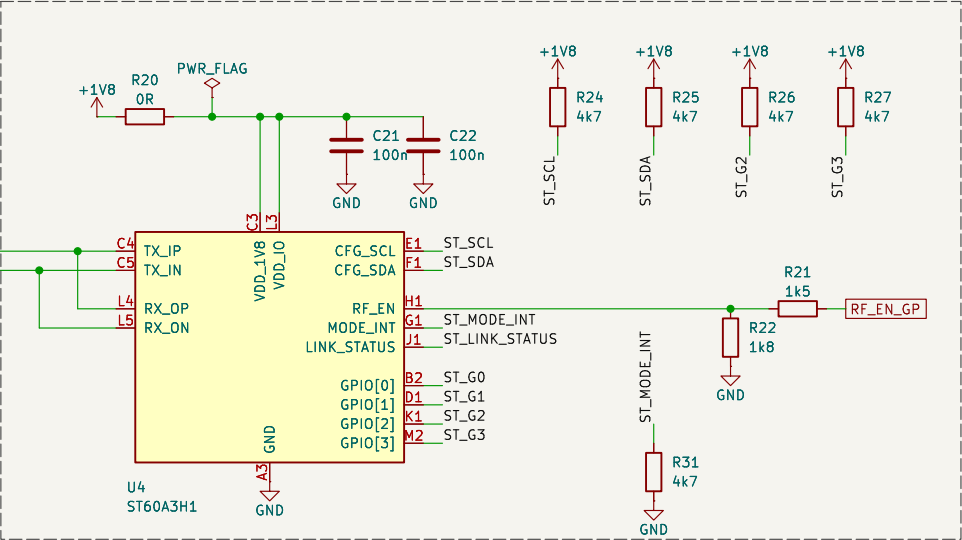
Header	Pin	Signal
J2 (LEFT)	1	+1V8
	2	GND
	3	TUN_G0
	4	TUN_G1
	5	CFG_SDA
	6	CFG_SCL
	7	GP21
	8	GP22
	9	GP23
	10	GP24
	11	GP26
	12	GP27
	13	GP28
	14	GP29
	15	+3V3
	16	VBUS
J3 (RIGHT)	1	ST_INT
	2	ST_LINK
	3	TUN_G2
	4	TUN_G3
	5	RF_EN_GP
	6	GP7
	7	GP6
	8	GP5
	9	GP4
	10	GP3
	11	GP2
	12	GP1
	13	GP0
	14	GND
	15	+3V3
	16	VBUS

CERN-OHL-P		
Drawn by tools/schlayout.py from the netlist in tools/circuit.py		
Sequoia Hope Alexander		
Sheet: /01_mcu/		
File: 01_mcu.kicad_sch		
Title: comms – RP2350A + ST60A3H1 60 GHz contactless link – 01_mcu		
Size: A3	Date: 2026-10-02	Rev: A
KiCad E.D.A. 9.0.8		Id: 2/4

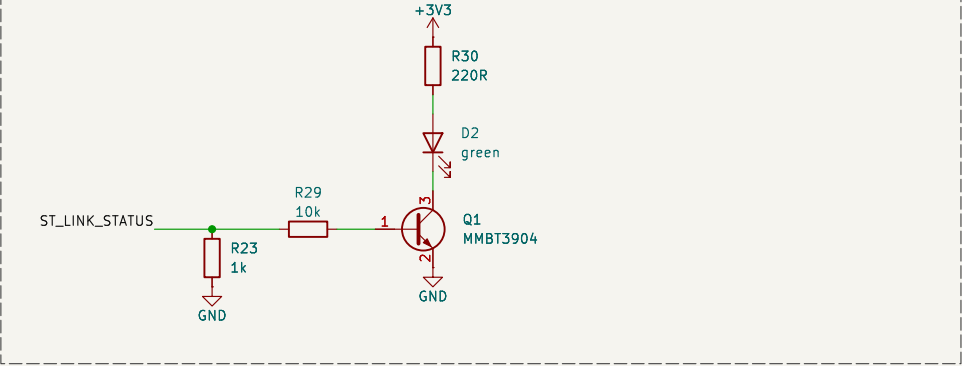
eUSB2 repeater



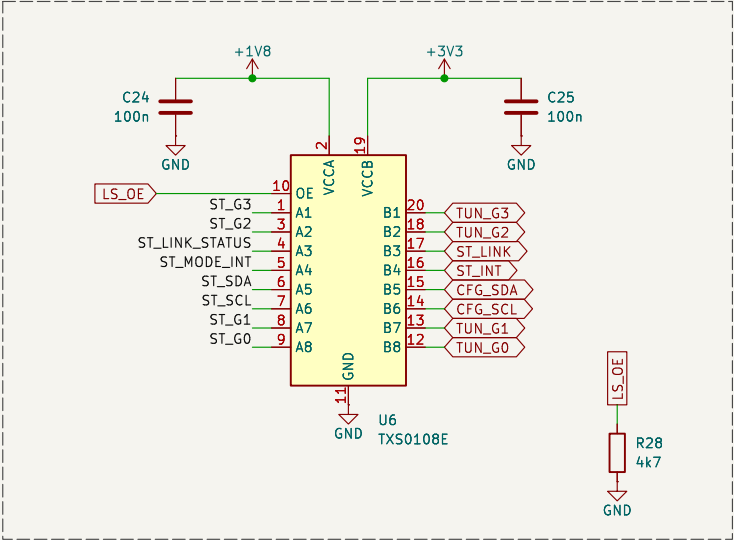
ST60A3H1



Link LED

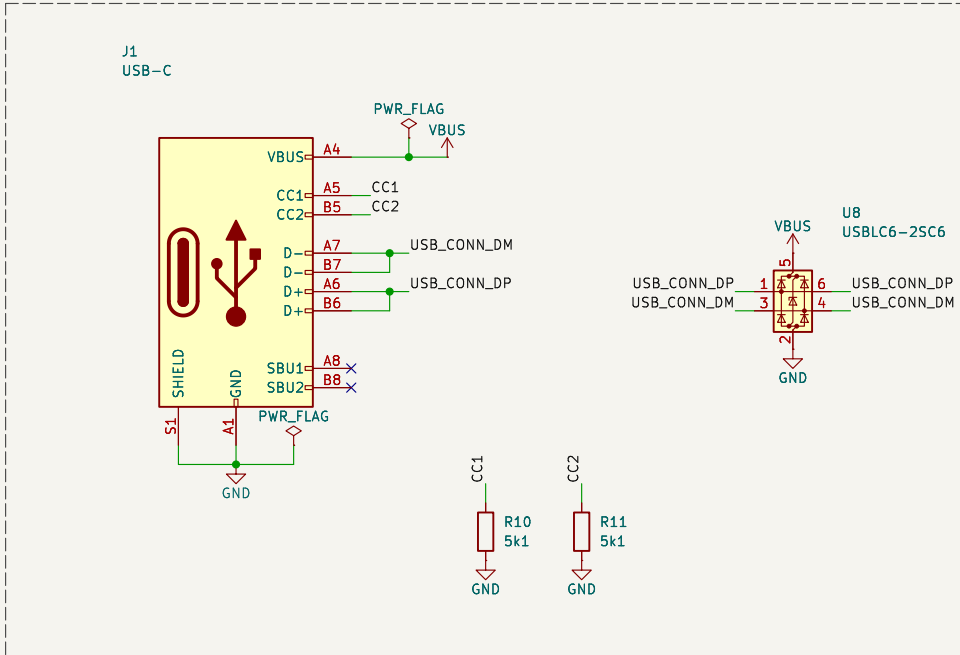


1.8 V – 3.3 V translator

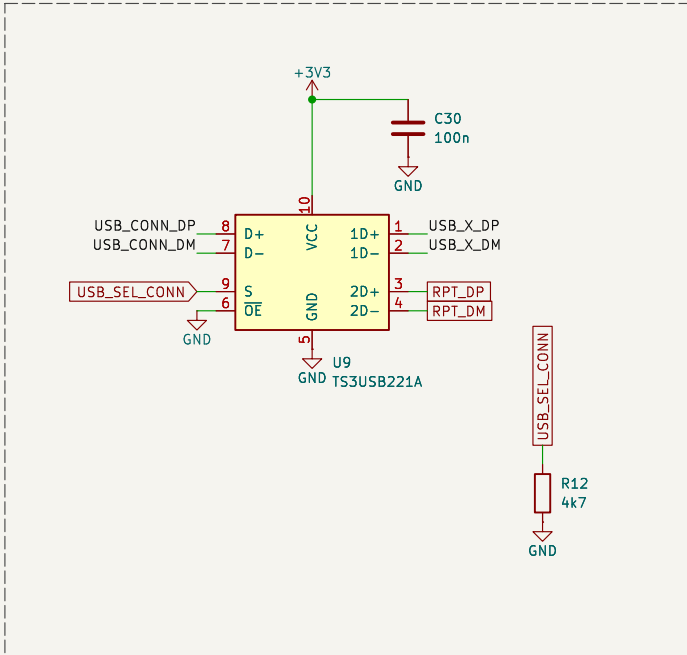


After ST's X-NUCLEO-60K1A1 (B2379A): the ST60A3H1 on +1V8 through a 0 R link, the PTN3222 on its eUSB2 pair and held in reset by LINK_STATUS, an eight-channel auto-direction translator to 3.3 V (TXS0108E here, ST2378E there), RF_EN by divider, the link LED off an NPN.
The translator is off (LS_OE low) until the RP2350 enables it, so the ST60 powers up with its own straps on its pins and R31 on MODE_INT. Pull-ups, 4k7 to 1.8 V, on the configuration bus and on GPIO[2]/[3], the tunnelled SCL/SDA; any added on the 3.3 V side (J2, J3) should be 8k2 or more.
Before turning +1V8 off: LS_OE low and RF_EN low. After turning it on: 2.25 ms before either goes high.
Configuration bus: ST60A3H1 at 0x60, PTN3222 at 0x43 (ADDR on +1V8).

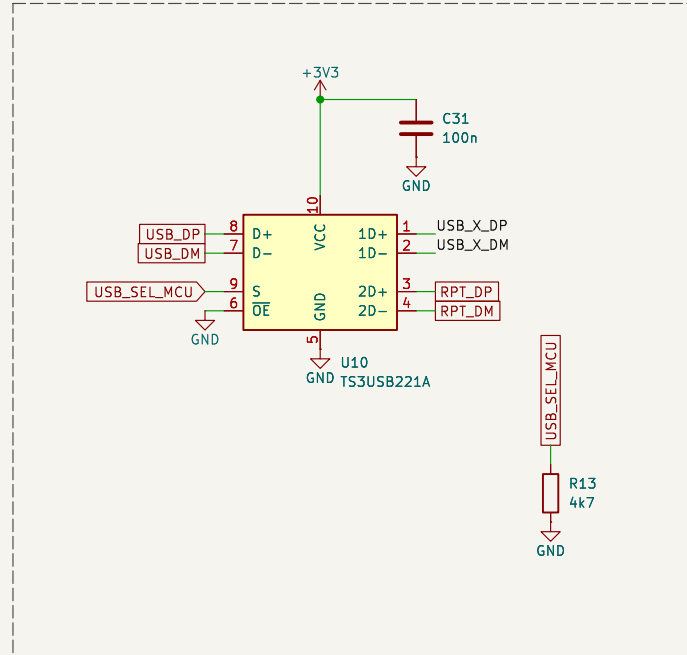
USB-C, device



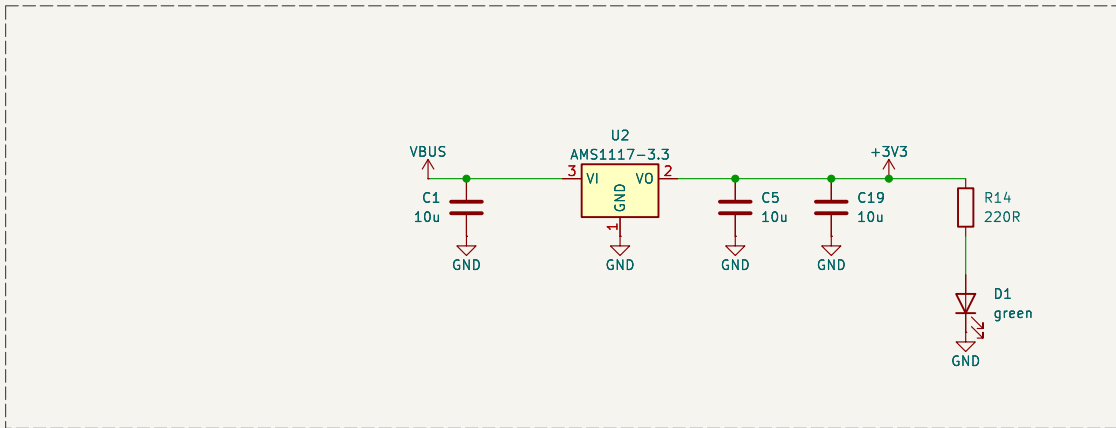
Switch: connector side



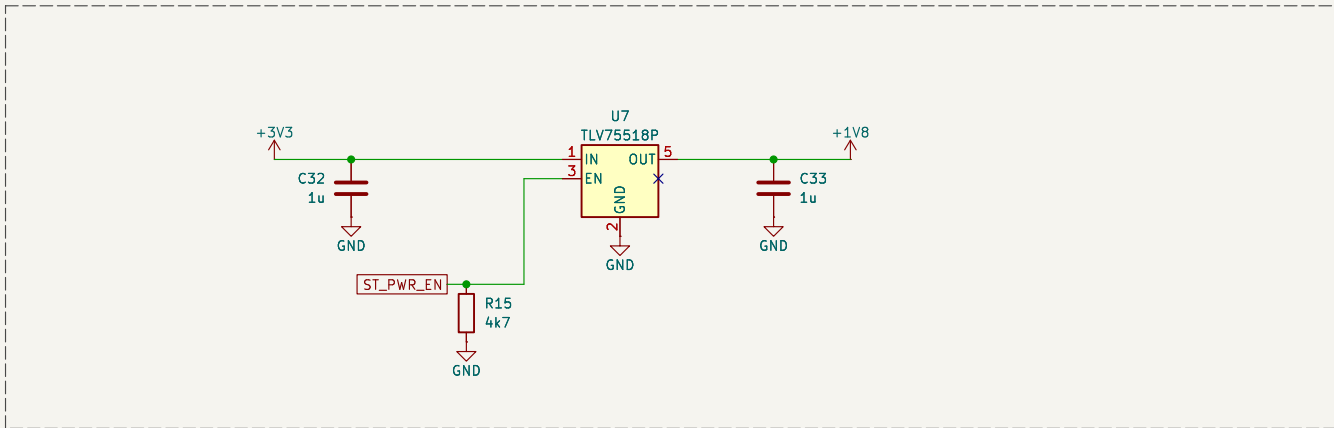
Switch: RP2350 side



3.3 V



1.8 V, switchable



Three things want the USB pair: the connector, the RP2350, the eUSB2 repeater. The two switches join any two through the node USB_SEL_CONN low, USB_SEL_MCU low: connector to RP2350 -- the board as it powers up, both selects pulled down.
CONN high: connector to repeater -- a host's USB goes over the air.
MCU high: RP2350 to repeater -- the RP2350 is the device at the far end.
Both high joins all three and is not a state to use.
ST_PWR_EN high turns the 1.8 V regulator on; it is pulled down, so the rail is off until the RP2350 asks for it. Off, the regulator discharges its output, which is a power-on reset for the ST60A3H1.